

Respiratory Physiology

Human Physiology - Chapter 16

Teaching Resources

December 24, 2025

Outline

16.1 The Respiratory System

16.2 Physical Aspects of Ventilation

16.3 Mechanics of Breathing

16.4 Gas Exchange in the Lungs

16.5 Regulation of Breathing

**16.6 Hemoglobin and Oxygen
Transport**

16.7 Carbon Dioxide Transport

16.8 Acid-Base Balance

16.9 Exercise and High Altitude

Outline

16.1 The Respiratory System

16.2 Physical Aspects of Ventilation

16.3 Mechanics of Breathing

16.4 Gas Exchange in the Lungs

16.5 Regulation of Breathing

16.6 Hemoglobin and Oxygen Transport

16.7 Carbon Dioxide Transport

16.8 Acid-Base Balance

16.9 Exercise and High Altitude

16.1 The Respiratory System

- The **respiratory system** (呼吸系统) performs **gas exchange** (气体交换)
- Delivers O_2 to tissues for **cellular respiration** (细胞呼吸) in mitochondria
- Removes CO_2 produced by metabolism
- **External respiration** (外呼吸): gas exchange between blood and environment

16.1 The Respiratory System

- **Alveoli** (肺泡) are microscopic air sacs where gas exchange occurs
- Approximately 300 million alveoli provide 50-100 m² surface area
- **Type I alveolar cells** (I 型肺泡细胞): thin cells facilitating diffusion (95% of surface)
- **Type II alveolar cells** (II 型肺泡细胞): secrete **pulmonary surfactant** (肺表面活性物质)
- **Pulmonary macrophages** (肺巨噬细胞): phagocytose inhaled particles



16.1 The Respiratory System

- The **blood-air barrier** (气血屏障) is extremely thin ($0.3\ \mu\text{m}$)
- Composed of: type I alveolar cell, fused **basement membranes** (基膜), and **capillary endothelium** (毛细血管内皮)
- Short diffusion distance enables rapid gas exchange
- Essential for efficient oxygen and carbon dioxide transfer



16.1 The Respiratory System

- **Alveolar pores** (肺泡孔) connect adjacent alveoli
- Provide **collateral ventilation** (侧支通气) if one airway becomes blocked
- Help equalize pressure between alveoli
- Contribute to efficient lung ventilation



16.1 The Respiratory System

- The respiratory system has two functional zones
- **Conducting zone** (传导区): airways that conduct air but no gas exchange
- Includes: nose, pharynx, larynx, trachea, bronchi, bronchioles, terminal bronchioles
- **Respiratory zone** (呼吸区): where gas exchange occurs
- Includes: respiratory bronchioles, alveolar ducts, alveolar sacs



16.1 The Respiratory System

- The **conducting zone** shows extensive branching pattern
- **Trachea** (气管) divides into right and left **primary bronchi** (主支气管)
- Successive branching creates the bronchial tree
- Each branch reduces in diameter but increases in total number
- Function: warm, humidify, and filter inspired air



16.1 The Respiratory System

- The **larynx** (喉) guards the entrance to the trachea
- Contains the **glottis** (声门): opening between **vocal folds** (声带)
- **Vocal folds** (true vocal cords): produce sound through vibration
- **Ventricular folds** (室带, false vocal cords): protect airway, no sound production
- Air must pass through glottis to enter or leave lungs



16.1 The Respiratory System

- The **thoracic cavity** (胸腔) contains the lungs
- **Pleural membranes** (胸膜) are double-layered serous membranes
- **Parietal pleura** (壁层胸膜): lines thoracic wall
- **Visceral pleura** (脏层胸膜): covers lung surface
- **Intrapleural space** (胸膜腔): potential space with thin fluid layer
- Fluid reduces friction and creates surface tension to keep lungs expanded



16.1 The Respiratory System

- **Radiographs** (X-rays) show normal lung position in thoracic cavity
- Lungs appear dark (radiolucent) because filled with air
- Heart visible in central **mediastinum** (纵隔)
- Ribs and other bony structures appear white (radiopaque)
- Useful for diagnosing pneumothorax, pneumonia, and other lung diseases



Outline

16.1 The Respiratory System

16.2 Physical Aspects of Ventilation

16.3 Mechanics of Breathing

16.4 Gas Exchange in the Lungs

16.5 Regulation of Breathing

16.6 Hemoglobin and Oxygen
Transport

16.7 Carbon Dioxide Transport

16.8 Acid-Base Balance

16.9 Exercise and High Altitude

16.2 Physical Aspects of Ventilation

- **Ventilation** (通气) is the mechanical process of breathing
- Air flows due to **pressure gradients** (压力梯度): from high to low pressure
- **Intrapulmonary pressure** (肺内压): pressure within alveoli
- **Intrapleural pressure** (胸膜腔内压): always subatmospheric (negative)
- Pressure changes drive inspiration and expiration

16.2 Physical Aspects of Ventilation

- Pressure changes during the respiratory cycle follow predictable patterns
- During **inspiration** (吸气): intrapulmonary pressure becomes subatmospheric
- During **expiration** (呼气): intrapulmonary pressure rises above atmospheric
- Intrapleural pressure remains negative throughout breathing cycle
- **Transpulmonary pressure** (跨肺压) keeps lungs inflated against chest wall



16.2 Physical Aspects of Ventilation

- **Pneumothorax** (气胸) occurs when air enters intrapleural space
- Intrapleural pressure equalizes with atmospheric pressure
- Loss of transpulmonary pressure causes lung collapse
- Ribs expand outward due to loss of elastic tension from collapsed lung
- Requires medical intervention to re-expand lung



16.2 Physical Aspects of Ventilation

- **Law of Laplace** (拉普拉斯定律): $P = \frac{2T}{r}$
- Surface tension creates pressure inversely proportional to radius
- Smaller alveoli would have higher pressure and collapse into larger ones
- Without **surfactant** (表面活性物质), breathing would be impossible



16.2 Physical Aspects of Ventilation

- **Pulmonary surfactant** is secreted by **type II alveolar cells** (II 型肺泡细胞)
- Composed of phospholipids (mainly **dipalmitoylphosphatidylcholine**) and proteins
- Reduces surface tension more in smaller alveoli
- Stabilizes all alveoli regardless of size
- Prevents **atelectasis** (肺不张, alveolar collapse)
- Reduces work of breathing by 50%



Outline

16.1 The Respiratory System

16.2 Physical Aspects of Ventilation

16.3 Mechanics of Breathing

16.4 Gas Exchange in the Lungs

16.5 Regulation of Breathing

16.6 Hemoglobin and Oxygen
Transport

16.7 Carbon Dioxide Transport

16.8 Acid-Base Balance

16.9 Exercise and High Altitude

16.3 Mechanics of Breathing

- **Boyle's law** (波义耳定律): pressure and volume are inversely related
- Increasing thoracic volume decreases pressure → inspiration
- Decreasing thoracic volume increases pressure → expiration
- The **diaphragm** (膈肌) is the primary muscle of breathing
- Innervated by **phrenic nerve** (膈神经, C3-C5 spinal nerves)

16.3 Mechanics of Breathing

- Lung volume changes are visible on radiographs
- During **expiration**: lungs are smaller, diaphragm domed upward
- During **inspiration**: lungs expand, diaphragm flattens downward
- Rib cage elevation also increases thoracic volume
- These volume changes create pressure gradients for airflow



16.3 Mechanics of Breathing

- **Inspiration muscles** (吸气肌):
 - Primary: **diaphragm** (膈肌) and **external intercostal muscles** (肋外肌)
 - Accessory: **sternocleidomastoid** (胸锁乳突肌) and **scalene muscles** (斜角肌)
- **Expiration muscles** (呼气肌):
 - Normal expiration: passive (elastic recoil)
 - Forced expiration: **internal intercostal muscles** (肋间内肌) and **abdominal muscles** (腹肌)



16.3 Mechanics of Breathing

- Normal quiet breathing uses mainly the diaphragm
- **Forced inspiration** (深呼吸): additional accessory muscles contract
- **Forced expiration**: active contraction of internal intercostals and abdominal muscles
- Comparison shows different muscle involvement for normal vs. forced breathing



16.3 Mechanics of Breathing

- **Mechanics of pulmonary ventilation**
shown at three time points
- Before inspiration: intrapulmonary pressure = atmospheric (equilibrium)
- During inspiration: volume increases
→ pressure decreases to 758 mmHg
→ air flows in
- During expiration: volume decreases
→ pressure increases to 762 mmHg
→ air flows out
- Demonstrates **Boyle's law** in action during breathing



16.3 Mechanics of Breathing

- **Spirometry** (肺量测定法) measures lung volumes and capacities
- **Lung volumes** (肺容量): **tidal volume** (潮气量), **inspiratory reserve** (补吸气量), **expiratory reserve** (补呼气量), **residual volume** (残气量)
- **Lung capacities** (肺量): sums of two or more volumes
- **Vital capacity** (肺活量): maximum air that can be exhaled after maximal inspiration
- **Total lung capacity** (肺总量): sum of all volumes



16.3 Mechanics of Breathing

- **Residual volume** cannot be measured by spirometry alone

16.3 Mechanics of Breathing

- Definitions of lung volumes and capacities are standardized
- **Tidal volume (TV)** (潮气量): 500 mL, normal breath
- **Vital capacity (VC)** (肺活量): 4600 mL, maximum expiration after maximum inspiration
- **Functional residual capacity (FRC)** (功能残气量): volume after normal expiration
- These measurements are essential for assessing respiratory function



16.3 Mechanics of Breathing

- **FEV₁** (一秒用力呼气量): forced expiratory volume in 1 second
- **FEV₁/FVC ratio**: percentage of vital capacity exhaled in first second
- Normal ratio: >80%
- Reduced ratio indicates **obstructive pulmonary disease** (阻塞性肺病)
- Examples: asthma, chronic bronchitis, emphysema
- Important diagnostic tool for respiratory disorders



16.3 Mechanics of Breathing

- **Ventilation terminology** defines key respiratory measurements
- **Minute ventilation** (分钟通气量): total air moved per minute = $TV \times$ respiratory rate
- **Anatomical dead space** (解剖死腔): conducting airways volume (150 mL)
- **Alveolar ventilation** (肺泡通气量): effective ventilation for gas exchange
- Alveolar ventilation = $(TV - \text{dead space}) \times$ respiratory rate



16.3 Mechanics of Breathing

- **Emphysema** (肺气肿) destroys alveolar walls
- Normal lung: numerous small alveoli with extensive surface area
- Emphysematous lung: fewer, larger air spaces
- Destruction reduces gas exchange surface area
- Loss of elastic recoil impairs expiration
- Usually caused by smoking or alpha-1 antitrypsin deficiency



Outline

16.1 The Respiratory System

16.2 Physical Aspects of Ventilation

16.3 Mechanics of Breathing

16.4 Gas Exchange in the Lungs

16.5 Regulation of Breathing

16.6 Hemoglobin and Oxygen
Transport

16.7 Carbon Dioxide Transport

16.8 Acid-Base Balance

16.9 Exercise and High Altitude

16.4 Gas Exchange in the Lungs

- **Atmospheric pressure** (大气压)
measured by mercury barometer
- At sea level: 760 mmHg (or 760 torr, or 1 atmosphere)
- Atmospheric pressure decreases with altitude
- This standard pressure is reference for all respiratory measurements
- Understanding pressure is essential for gas exchange principles



16.4 Gas Exchange in the Lungs

- At high altitude, atmospheric pressure decreases
- **Partial pressure of oxygen** (P_{O_2}) decreases proportionally
- At 10,000 feet: $P_{O_2} \approx 110$ mmHg (vs. 159 mmHg at sea level)
- Reduced P_{O_2} causes **hypoxia** (缺氧)
- Triggers **acclimatization** (适应) responses



16.4 Gas Exchange in the Lungs

- **Partial pressures** (分压) of gases change from inspired air to alveolar air
- Inspired air: $P_{O_2} = 159$ mmHg,
 $P_{CO_2} = 0.3$ mmHg
- Alveolar air: $P_{O_2} = 105$ mmHg,
 $P_{CO_2} = 40$ mmHg
- O_2 decreases due to uptake into blood
- CO_2 increases due to diffusion from blood
- Alveolar air is saturated with water vapor (47 mmHg at 37°C)



16.4 Gas Exchange in the Lungs

- Extensive contact between **pulmonary capillaries** (肺毛细血管) and alveoli
- Large surface area (50-100 m²) maximizes gas exchange
- Thin barrier (0.3 μm) minimizes diffusion distance
- Each alveolus surrounded by dense capillary network
- Enables rapid and efficient gas exchange



16.4 Gas Exchange in the Lungs

- **Blood gas measurements** use specialized electrodes
- **P_{O₂} electrode** measures oxygen partial pressure in blood
- Electrode first calibrated with solutions of known oxygen concentration
- Then inserted into blood sample for measurement
- Essential clinical tool for assessing respiratory function and oxygenation



16.4 Gas Exchange in the Lungs

- **Partial pressures** of gases in blood during circulation
- Deoxygenated blood entering lungs:
 $P_{O_2} = 40 \text{ mmHg}$, $P_{CO_2} = 46 \text{ mmHg}$
- Oxygenated blood leaving lungs:
 $P_{O_2} = 100 \text{ mmHg}$, $P_{CO_2} = 40 \text{ mmHg}$
- Gases diffuse down their partial pressure gradients
- Systemic capillaries: O_2 diffuses into tissues, CO_2 diffuses into blood



16.4 Gas Exchange in the Lungs

- **Ventilation-perfusion ratio** (V/Q ratio, 通气血流比) varies in different lung regions
- Apex of lungs: $V/Q > 1$ (relatively overventilated)
- Base of lungs: $V/Q < 1$ (relatively underventilated)
- Regional variation affects gas exchange efficiency
- **V/Q mismatch** (通气血流失衡) impairs gas exchange in disease



Outline

16.1 The Respiratory System

16.2 Physical Aspects of Ventilation

16.3 Mechanics of Breathing

16.4 Gas Exchange in the Lungs

16.5 Regulation of Breathing

16.6 Hemoglobin and Oxygen
Transport

16.7 Carbon Dioxide Transport

16.8 Acid-Base Balance

16.9 Exercise and High Altitude

16.5 Regulation of Breathing

- Breathing controlled by **respiratory centers** (呼吸中枢) in brain stem
- **Medullary rhythmicity center** (延髓节律中枢): generates basic rhythm
- **Dorsal respiratory group (DRG)**: controls inspiration
- **Ventral respiratory group (VRG)**: controls forced expiration
- **Pontine respiratory centers** (脑桥呼吸中枢):
 - **Pneumotaxic center**: inhibits inspiration



16.5 Regulation of Breathing

- **Apneustic center:** stimulates inspiration

16.5 Regulation of Breathing

- **Peripheral chemoreceptors** (外周化学感受器) monitor blood gases
- **Carotid bodies** (颈动脉体) in carotid arteries
- **Aortic bodies** (主动脉体) in aortic arch
- Detect changes in arterial P_{O_2} , P_{CO_2} , and pH
- Send signals via **glossopharyngeal nerve** (IX) and **vagus nerve** (X)
- Provide rapid feedback to brain stem respiratory centers



16.5 Regulation of Breathing

- Flowchart shows negative feedback control of ventilation
- Multiple inputs converge on medullary rhythmicity center
- **Central chemoreceptors** (中枢化学感受器): respond to CO_2 /pH changes
- **Peripheral chemoreceptors**: respond to O_2 , CO_2 , pH
- **Higher brain centers**: voluntary control, emotions
- Motor output to respiratory muscles controls breathing



16.5 Regulation of Breathing

- Inverse relationship between **minute ventilation** (分钟通气量) and arterial P_{CO_2}
- Doubling ventilation halves arterial P_{CO_2}
- Demonstrates the importance of ventilation in CO_2 regulation
- CO_2 is tightly controlled to maintain blood pH
- Small changes in P_{CO_2} significantly affect ventilation



16.5 Regulation of Breathing

- **Chemoreceptor control** showing negative feedback loop
- Increased blood CO_2 crosses **blood-brain barrier** (血脑屏障)
- Forms H^+ ions in **cerebrospinal fluid** (脑脊液)
- H^+ stimulates **central chemoreceptors** in medulla
- Increases ventilation $\rightarrow$ lowers blood $\text{CO}_2 \rightarrow$ completes feedback loop



16.5 Regulation of Breathing

- Mechanism of how blood CO_2 affects central chemoreceptors
- CO_2 diffuses across blood-brain barrier into CSF
- **Carbonic anhydrase** (碳酸酐酶)
catalyzes: $\text{CO}_2 + \text{H}_2\text{O} \longrightarrow \text{H}_2\text{CO}_3$
 $\longrightarrow \text{H}^+ + \text{HCO}_3^-$
- Increased H^+ in CSF stimulates central chemoreceptors
- This indirect mechanism explains CO_2 's strong effect on breathing



16.5 Regulation of Breathing

- Comparison of CO_2 and O_2 effects on breathing
- Increased blood CO_2 : steep curve, strong stimulus for ventilation
- Decreased blood P_{O_2} : shallow curve until $\text{P}_{\text{O}_2} < 60 \text{ mmHg}$
- CO_2 is the primary regulator of ventilation under normal conditions
- O_2 becomes important stimulus only during significant **hypoxemia** (低氧血症)



16.5 Regulation of Breathing

- Sensitivity of chemoreceptors to different stimuli
- **Central chemoreceptors:** primarily respond to CO_2 (via H^+ changes in CSF)
- **Peripheral chemoreceptors:** respond to O_2 , CO_2 , and pH
- Carotid bodies more sensitive to O_2 than aortic bodies
- Table shows relative importance of each stimulus for each receptor type



Outline

16.1 The Respiratory System

16.2 Physical Aspects of Ventilation

16.3 Mechanics of Breathing

16.4 Gas Exchange in the Lungs

16.5 Regulation of Breathing

**16.6 Hemoglobin and Oxygen
Transport**

16.7 Carbon Dioxide Transport

16.8 Acid-Base Balance

16.9 Exercise and High Altitude

16.6 Hemoglobin and Oxygen Transport

- Most O_2 (98.5%) transported bound to **hemoglobin** (血红蛋白, Hb)
- Only 1.5% of O_2 dissolved in plasma
- Hemoglobin dramatically increases blood oxygen-carrying capacity
- Without hemoglobin, cardiac output would need to be 80 L/min (vs. normal 5 L/min)



16.6 Hemoglobin and Oxygen Transport

- **Hemoglobin** structure: quaternary protein
- Four polypeptide chains: 2 alpha chains, 2 beta chains
- Four **heme groups** (血红素), each with Fe^{2+} that binds one O_2
- One Hb molecule binds 4 O_2 molecules
- **Oxyhemoglobin (HbO_2)** (氧合血红蛋白): $\text{Hb} + \text{O}_2$, bright red
- **Deoxyhemoglobin** (脱氧血红蛋白): Hb without O_2 , dark red



16.6 Hemoglobin and Oxygen Transport

- Numerical relationship between P_{O_2} and Hb saturation at pH 7.40 and 37°C
- At $P_{O_2} = 100$ mmHg (arterial): 97% saturation
- At $P_{O_2} = 40$ mmHg (venous): 75% saturation
- 22% of O_2 unloaded to tissues during rest
- During exercise, more O_2 can be unloaded as tissue P_{O_2} decreases



16.6 Hemoglobin and Oxygen Transport

- **Oxyhemoglobin dissociation curve** (氧合血红蛋白解离曲线) is S-shaped (sigmoid)
- Shape reflects **cooperative binding** (协同结合)
- First O_2 binding facilitates subsequent O_2 binding
- Steep middle portion: efficient O_2 loading in lungs and unloading in tissues
- Plateau at high P_{O_2} : safety margin for arterial oxygenation



16.6 Hemoglobin and Oxygen Transport

- Effect of pH on hemoglobin affinity and oxygen unloading
- Lower pH (higher H^+): decreased affinity, more O_2 unloaded
- **Bohr effect** (波尔效应): physiological adaptation
- Metabolically active tissues produce CO_2 and H^+
- Lower pH in these tissues enhances O_2 delivery where needed



16.6 Hemoglobin and Oxygen Transport

- **Bohr effect** shown graphically
- Decreased pH shifts curve rightward (decreased affinity)
- At any given P_{O_2} , lower saturation means more O_2 released
- Adaptation helps deliver more O_2 to active tissues producing acids
- Example: exercising muscle has lower pH, receives more O_2



16.6 Hemoglobin and Oxygen Transport

- **2,3-DPG** (2,3-diphosphoglycerate, 2,3-二磷酸甘油酸) promotes O_2 unloading
- Binds to deoxyhemoglobin, stabilizing it
- Production inhibited by oxyhemoglobin
- At high altitude: low $P_{O_2} \rightarrow$ less oxyhemoglobin $\rightarrow$ more 2,3-DPG
- Shifts curve rightward, enhances O_2 unloading to compensate for hypoxia



16.6 Hemoglobin and Oxygen Transport

- Summary of factors affecting hemoglobin-oxygen affinity
- **Rightward shift** (decreased affinity, enhanced unloading):
 - Increased temperature, increased P_{CO_2} , decreased pH, increased 2,3-DPG
- **Leftward shift** (increased affinity, reduced unloading):
 - Decreased temperature, decreased P_{CO_2} , increased pH, decreased 2,3-DPG



16.6 Hemoglobin and Oxygen Transport

- These adaptations optimize O_2 delivery to tissues

16.6 Hemoglobin and Oxygen Transport

- **Sickle-cell anemia** (镰状细胞贫血):
genetic hemoglobin disorder
- Mutation in beta chain causes Hb polymerization at low O_2
- Red blood cells deform into crescent (sickle) shape
- Sickled cells are rigid, block capillaries, cause pain crises
- Cells are fragile, leading to hemolytic anemia



16.6 Hemoglobin and Oxygen Transport

- Comparison of **hemoglobin** and **myoglobin** (肌红蛋白) dissociation curves
- Myoglobin: oxygen-binding protein in skeletal muscle
- Hyperbolic curve (not sigmoid), much higher O_2 affinity than Hb
- At low P_{O_2} in exercising muscle: Hb releases O_2 , myoglobin retains it
- Myoglobin serves as O_2 reservoir in muscle tissue



Outline

16.1 The Respiratory System

16.2 Physical Aspects of Ventilation

16.3 Mechanics of Breathing

16.4 Gas Exchange in the Lungs

16.5 Regulation of Breathing

16.6 Hemoglobin and Oxygen
Transport

16.7 Carbon Dioxide Transport

16.8 Acid-Base Balance

16.9 Exercise and High Altitude

16.7 Carbon Dioxide Transport

- CO₂ transported in three forms:
 - Dissolved CO₂ (7%): directly dissolved in plasma
 - **Carbaminohemoglobin** (氨基甲酸血红蛋白, 23%): CO₂ bound to Hb amino groups
 - **Bicarbonate** (HCO₃⁻, 70%): most abundant form
- **Chloride shift** (氯移位): Cl⁻ enters RBCs as HCO₃⁻ exits
- Maintains electrical neutrality across RBC membrane



16.7 Carbon Dioxide Transport

- In lungs, **reverse chloride shift** (反向氯移位) occurs
- HCO_3^- re-enters RBCs, combines with H^+
- Forms H_2CO_3 , then $\text{CO}_2 + \text{H}_2\text{O}$
- CO_2 diffuses into alveoli for exhalation
- Cl^- exits RBCs back into plasma
- Process facilitated by oxygenation of Hb (**Haldane effect**, 霍尔丹效应)



Outline

16.1 The Respiratory System

16.2 Physical Aspects of Ventilation

16.3 Mechanics of Breathing

16.4 Gas Exchange in the Lungs

16.5 Regulation of Breathing

16.6 Hemoglobin and Oxygen
Transport

16.7 Carbon Dioxide Transport

16.8 Acid-Base Balance

16.9 Exercise and High Altitude

16.8 Acid-Base Balance

- Normal blood pH: 7.35-7.45
- **Bicarbonate buffer system:** $\text{H}_2\text{CO}_3 \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$
- HCO_3^- neutralizes excess H^+ ions
- Maintains pH within narrow physiological range
- Essential for proper enzyme function and metabolism



16.8 Acid-Base Balance

- Terminology for acid-base balance
- **Acidosis** (酸中毒): $\text{pH} < 7.35$
 - **Respiratory acidosis**: due to increased P_{CO_2}
 - **Metabolic acidosis**: due to increased metabolic acids or loss of HCO_3^-
- **Alkalosis** (碱中毒): $\text{pH} > 7.45$
 - **Respiratory alkalosis**: due to decreased P_{CO_2}
 - **Metabolic alkalosis**: due to loss of H^+ or gain of HCO_3^-



16.8 Acid-Base Balance

- Classification of acid-base disorders
- Primary disorder identified by pH direction and compensatory changes
- Respiratory disorders: primary change in P_{CO_2}
- Metabolic disorders: primary change in HCO_3^-
- Compensation: body attempts to restore pH toward normal



16.8 Acid-Base Balance

- Effect of lung function on acid-base balance
- **Hypoventilation** (通气不足):
increased P_{CO_2} → respiratory acidosis
- **Hyperventilation** (通气过度):
decreased P_{CO_2} → respiratory alkalosis
- Demonstrates lungs' crucial role in pH regulation
- Respiratory compensation occurs within minutes to hours



Outline

16.1 The Respiratory System

16.2 Physical Aspects of Ventilation

16.3 Mechanics of Breathing

16.4 Gas Exchange in the Lungs

16.5 Regulation of Breathing

16.6 Hemoglobin and Oxygen
Transport

16.7 Carbon Dioxide Transport

16.8 Acid-Base Balance

16.9 Exercise and High Altitude

16.9 Exercise and High Altitude

- During moderate exercise, arterial blood gases remain remarkably stable
- P_{O_2} , P_{CO_2} , and pH show minimal changes
- Only during very heavy exercise does P_{CO_2} slightly change
- Demonstrates body's ability to maintain homeostasis during increased metabolic demand
- Precise matching of ventilation to metabolism



16.9 Exercise and High Altitude

- Exercise increases respiratory function proportionally to metabolic rate
- **Tidal volume** increases (up to 50% of vital capacity)
- **Breathing rate** increases
- **Minute ventilation** increases dramatically
- **Pulmonary blood flow** increases
- Despite these changes, arterial blood gases remain stable



16.9 Exercise and High Altitude

- At high altitude, atmospheric pressure and P_{O_2} decrease
- Arterial P_{O_2} and oxyhemoglobin saturation decrease
- Results in **hypoxia** (缺氧)
- Example: At 10,000 feet, arterial P_{O_2} drops to 67 mmHg (vs. 100 mmHg at sea level)
- Triggers multiple **acclimatization** (适应) responses



16.9 Exercise and High Altitude

- **Acclimatization** involves multiple adaptations over different time scales
- Immediate (minutes):
hyperventilation in response to low P_{O_2}
- Hours-days: increased 2,3-DPG production
- Days-weeks: increased RBC production via **erythropoietin** (促红细胞生成素)
- Weeks-months: increased capillary density in tissues



16.9 Exercise and High Altitude

- These adaptations improve O_2 delivery despite reduced P_{O_2}

16.9 Exercise and High Altitude

- Timeline of respiratory adaptations to high altitude
- ① Immediate: hyperventilation (peripheral chemoreceptor response)
- ② Hours: increased 2,3-DPG shifts oxyhemoglobin curve
- ③ Days: kidneys secrete erythropoietin
- ④ Weeks: increased RBC production (polycythemia)
- ⑤ Months: increased capillary and mitochondrial density



16.9 Exercise and High Altitude

- These sequential adaptations enable life at high altitude